# MANIPAL INSTITUTE OF TECHNOLOGY

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*(A constituent unit of MAHE, Manipal)*

## Mini Project Report
## of
## Database Management System Lab (MCA 4141)

# TOY MANUFACTURING AND IMPORT DEPENDENCY ANALYSIS SYSTEM


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# CERTIFICATE

This is to certify that the project titled "Toy Manufacturing and Import Dependency Analysis System" is a record of the bona fide work done by Student(s) (Reg. No. _____) submitted in partial fulfilment of the requirements for the award of the Degree of Master of Computer Applications of Manipal Institute of Technology, Manipal, Karnataka, during the academic year 2026–2027.

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# TABLE OF CONTENTS


| SECTION | PAGE |
|---|---|
| ABSTRACT | |
| CHAPTER 1: INTRODUCTION | |
| CHAPTER 2: PROBLEM STATEMENT & OBJECTIVES | |
| CHAPTER 3: FUNCTIONAL REQUIREMENTS – MODULES AND SOFTWARE USED | |
| CHAPTER 4: ER DIAGRAM & RELATIONAL TABLES ALONG WITH SAMPLE DATA | |
| CHAPTER 5: DDL COMMANDS & PL/SQL PROCEDURES / FUNCTIONS / TRIGGERS | |
| CHAPTER 6: RESULTS & SNAPSHOTS | |
| CHAPTER 7: CONCLUSION, LIMITATIONS & FUTURE WORK | |
| CHAPTER 8: REFERENCES | |


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# ABSTRACT

The Toy Manufacturing and Import Dependency Analysis System is a database-oriented mini project designed to manage and analyze the lifecycle of toy manufacturing activities. The system maintains structured information about toys, raw materials, suppliers, sourcing options, manufacturing batches, inventory, sales channels, and sales transactions. The proposed database supports the operational flow from raw material sourcing to toy manufacturing, inventory maintenance, sales, and profit analysis.

The distinctive feature of the proposed system is its Import Dependency and Local Substitution Impact Analysis. The system identifies raw materials sourced through imported suppliers, determines whether locally available alternatives exist, and compares their costs. This allows the manufacturer to evaluate the possible impact of local substitution on manufacturing cost and potential profitability. The system does not assume that local sourcing is always cheaper; instead, it provides a data-driven comparison.

The project is intended for implementation using Oracle SQL and PL/SQL and demonstrates relational database design, normalization, constraints, joins, aggregate functions, nested queries, views, indexes, sequences, stored procedures, functions, and triggers. The scope is intentionally limited so that it remains a feasible DBMS mini project rather than a complete enterprise resource planning system.

# CHAPTER 1: INTRODUCTION

## 1.1 Introduction

Toy manufacturing involves interconnected activities such as raw material procurement, supplier selection, production, stock management, and sales. When these activities are maintained through separate records, it becomes difficult to obtain a consolidated view of production cost, sourcing information, inventory, and sales performance.

## 1.2 Interpretation of the Original Abstract

The original abstract requires a system that can provide information about toys, imported sources, raw materials, manufacturing stages, e-commerce sales, profit analysis, and reduction of imported goods. If interpreted as a complete business application, this would require supply-chain management, procurement, logistics, accounting, and external platform integration. Such a system would be too large for a DBMS mini project. Therefore, the proposed project refines the abstract into a focused relational database and decision-support system.

## 1.3 Need for the Proposed System

- Centralized storage of toy, material, supplier, production, inventory, and sales data.
- Traceability from raw material sourcing to finished toy sales.
- Identification of imported and local sourcing options.
- Manufacturing cost and profit analysis through SQL queries.
- A practical method to analyze local alternatives for imported materials.

## 1.4 Scope of the Project

The project is organized into four integrated modules: (1) Toy & Material Management, (2) Supplier & Sourcing Management, (3) Manufacturing & Inventory Management, and (4) Sales & Business Analysis. Together, these modules cover raw material sourcing, toy requirements, production batches, inventory, sales, profit analysis, and import dependency analysis. The project excludes real-time e-commerce APIs, online payments, customer management, logistics tracking, customs processing, demand forecasting, machine learning, and complete ERP functionality.

# CHAPTER 2: PROBLEM STATEMENT & OBJECTIVES

## 2.1 Problem Statement

Toy manufacturers require a structured mechanism for maintaining information about toys, raw materials, suppliers, manufacturing activities, inventory, and sales. Dependence on imported raw materials may influence production cost and sourcing decisions. When sourcing information is not integrated with toy requirements and manufacturing records, it becomes difficult to determine which toys depend on imported materials, whether local alternatives exist, and how a sourcing change may affect cost and profit.

## 2.2 Objectives

1. **Module 1 – Toy & Material Management:**
   Maintain information about toys, raw materials, and the material requirements for each toy.
2. **Module 2 – Supplier & Sourcing Management:**
   Maintain supplier information and classify available raw material sourcing options as **LOCAL** or **IMPORTED**, including country of origin, cost, and availability.
3. **Module 2 – Supplier & Sourcing Management:**
   Identify imported raw materials and analyze the availability of suitable local sourcing alternatives.
4. **Module 3 – Manufacturing & Inventory Management:**
   Track manufacturing batches, actual material usage, production costs, and finished toy inventory.
5. **Module 4 – Sales & Business Analysis:**
   Record sales through different sales channels and calculate toy-wise revenue, manufacturing cost, and profit.
6. **Module 4 – Sales & Business Analysis:**
   Compare imported and local sourcing costs and evaluate their potential impact on manufacturing cost and profitability.

## 2.3 Existing System

A simplified existing approach may rely on spreadsheets, manual registers, and disconnected files. Supplier data, production records, stock, and sales information may be maintained separately. This can lead to data redundancy, inconsistent updates, difficulty in tracing relationships, and manual calculation of cost and profitability.

## 2.4 Proposed System

The proposed system integrates the manufacturing lifecycle through four consolidated database modules rather than treating every business activity as a separate module. Relational tables and foreign keys connect suppliers, raw materials, toys, manufacturing batches, inventory, sales channels, and sales transactions.

## 2.5 Novelty / Unique Feature

The Import Dependency and Local Substitution Impact Analysis is the project's principal novelty. A raw material can have multiple supplier-material records. Some may be imported and others local. The system compares available sourcing options for the same material and estimates the resulting cost difference when multiplied by the quantity required for a toy. This makes the project a database-based decision-support system rather than only an inventory or manufacturing management system.

# CHAPTER 3: FUNCTIONAL REQUIREMENTS – MODULES AND SOFTWARE USED

## 3.1 Modules of the System

| MODULE | OBJECTIVE | TABLES | OUTPUT |
|---|---|---|---|
| 1. Toy & Material Management | **Objective 1:** Maintain information about toys, raw materials, and the material requirements for each toy. | TOY, RAW_MATERIAL, TOY_MATERIAL | Toy master list, material master list, and toy-wise material requirement report |
| 2. Supplier & Sourcing Management | **Objective 2:** Maintain supplier information and classify available raw material sourcing options as **LOCAL** or **IMPORTED**, including country of origin, cost, and availability.<br><br>**Objective 3:** Identify imported raw materials and analyze the availability of suitable local sourcing alternatives. | SUPPLIER, SUPPLIER_MATERIAL | Supplier-wise supply report, imported versus local sourcing report, and local substitution comparison |
| 3. Manufacturing & Inventory Management | **Objective 4:** Track manufacturing batches, actual material usage, production costs, and finished toy inventory. | MANUFACTURING_BATCH, BATCH_MATERIAL_USAGE, INVENTORY | Manufacturing history, batch cost report, current inventory, and low-stock report |
| 4. Sales & Profit Analysis | **Objective 5:** Record sales through different sales channels and calculate toy-wise revenue, manufacturing cost, and profit.<br><br>**Objective 6:** Compare imported and local sourcing costs and evaluate their potential impact on manufacturing cost and profitability. | SALES_CHANNEL, SALE (with analytical use of related tables) | Channel-wise sales, toy-wise revenue, profit analysis, most profitable toys, and import dependency impact reports |

## 3.2 Functional Requirements

Module 1 – Toy & Material Management

- Add, update, and retrieve toy and raw material records.
- Define and maintain the raw material requirements for each toy through a bill-of-materials relationship.

Module 2 – Supplier & Sourcing Management

- Add and maintain supplier information and supplier-material sourcing options.
- Classify raw material sources as **LOCAL** or **IMPORTED** and record country of origin, unit cost, and availability.
- Identify imported raw materials and available local sourcing alternatives.

Module 3 – Manufacturing & Inventory Management

- Record manufacturing batches, actual material usage, labor cost, overhead cost, and production status.
- Maintain and update finished toy inventory based on manufacturing and sales activities.

Module 4 – Sales & Business Analysis

- Record sales transactions through different sales channels.
- Generate toy-wise revenue, manufacturing cost, and profit reports.
- Compare imported and local sourcing costs and analyze their potential impact on manufacturing cost and profitability.

## 3.3 Non-Functional Requirements

- Data integrity through primary keys, foreign keys, and CHECK constraints.
- Normalized design to reduce redundancy.
- Implementation using Oracle SQL and PL/SQL.
- Simple and maintainable structure suitable for academic demonstration.
- No dependence on external APIs or advanced infrastructure.

## 3.4 Hardware and Software Requirements

| CATEGORY | REQUIREMENT |
|---|---|
| Operating System | Windows or another OS supported by Oracle development tools |
| Database | Oracle Database |
| Development Tool | Oracle SQL Developer or equivalent |
| Languages | SQL and PL/SQL |
| Documentation | Microsoft Word or compatible software |
| Hardware | Standard laptop/desktop capable of running Oracle development tools |

## 3.5 Overall System Workflow

**Supplier / Source → Raw Material → Toy Material Requirement → Manufacturing Batch → Inventory → Sales Channel and Sale → Profit Analysis.** The **Import Dependency and Local Substitution Impact Analysis** is integrated across **Module 2 (Supplier & Sourcing Management)** and **Module 4 (Sales & Business Analysis)**. Module 2 maintains information about imported and locally sourced materials and identifies available local alternatives, while Module 4 uses this sourcing information along with manufacturing and sales data to evaluate the potential impact of sourcing decisions on manufacturing cost and profitability.